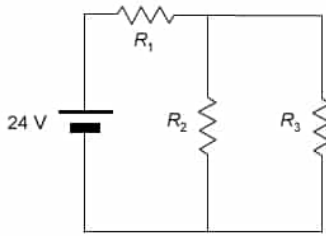


In the circuit shown below the currents through resistors R_2 and R_3 are 4 A and 2 A, respectively. Resistor R_1 has a resistance of 1.33 Ω .



What is the voltage drop across resistor R_3 ?

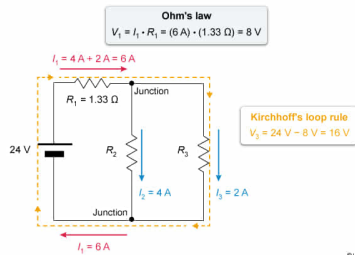
- ☐ A. 24 V [18%]
☐ B. 20 V [13%]
☐ C. 18 V [32%]
☒ D. 16 V [35%]

Correct

35% answered correctly

Explanation:

Voltage drop (V_1) across R_1 from Kirchhoff's rules and Ohm's law



Electrical circuits exhibit conservation of electrical charge (ie, no charge is lost in the circuit). Consequently, [Kirchhoff's junction rule](#) states that the total current I_{enter} entering a junction must equal the total current I_{exit} exiting the junction:

$$I_{\text{enter}} = I_{\text{exit}}$$

Electrical circuits also exhibit conservation of energy around a *closed path* (loop) in the circuit, in which the direction of the path matches the direction of the current in the loop. As a result, **Kirchhoff's loop rule** states that the sum of the voltage drops V around any closed loop in a circuit equals zero:

$$\sum_{i=1}^n V_i = V_1 + V_2 + \dots + V_n = 0$$

Together with [Ohm's law](#), Kirchhoff's rules can be used to solve for unknown variables in a circuit configuration.

In this circuit, the junction rule implies the current I_1 through resistor R_1 is equal to the sum of the given currents through resistors R_2 and R_3 :

$$I_1 = 4 \text{ A} + 2 \text{ A} = 6 \text{ A}$$

Using Ohm's law, the voltage drop V_1 across resistor R_1 is calculated to be:

$$V_1 = I_1 R_1 = (6 \text{ A})(1.33 \text{ } \Omega) = 8 \text{ V}$$

Applying the loop rule, the positive voltage V_{battery} across the battery *minus* the voltage drops V_1 and V_3 (across resistors R_1 and R_3) must be zero:

$$V_{\text{battery}} - V_1 - V_3 = 0$$

Solving for V_3 yields:

$$V_3 = V_{\text{battery}} - V_1$$

$$V_3 = 24 \text{ V} - 8 \text{ V} = 16 \text{ V}$$

Therefore, the voltage drop across resistor R_3 is 16 V.

(Choice A) Resistor R_3 is not in parallel with the battery, so R_3 cannot have the same voltage as the battery.

(Choice B) Incorrectly assuming the current through R_1 equals the average of the currents through R_2 and R_3 yields a calculated voltage of 20 V across R_3 .

(Choice C) Mistaking the current through R_1 as the voltage across R_1 yields a calculated voltage of 18 V across R_3 .

Educational objective:

Unknown currents in different branches of a circuit can be found by applying the junction rule, which states that total current entering a junction must equal the total current exiting the junction. Ohm's law and the loop rule are used to determine voltage and resistor values.

Time spent: 0 sec

Subject:
Physics

QID: 404043

System:
Electrostatics and Circuits

A researcher is working with a circuit that consists of a 10 V battery and two parallel resistors with an equivalent resistance of around 2 MΩ. The researcher measures the current in each resistor to be $2.23 \pm 0.04 \mu\text{A}$ and $2.75 \pm 0.05 \mu\text{A}$, respectively. Should the researcher accept the value of these currents?

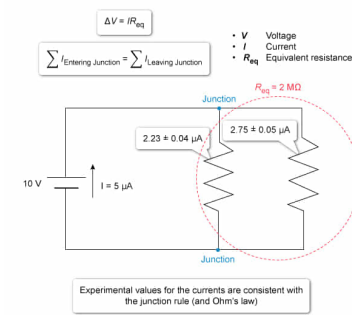
- ✓ ☒ A. Yes, because they are perfectly acceptable values. [27%]
- ☐ B. No, because the values are much smaller than the current flowing through the battery. [29%]
- ☐ C. No, because the values are not consistent with conservation of charge. [35%]
- ☐ D. No, because the uncertainty is more than 1% of the value. [7%]

Correct

27% answered correctly

Explanation:

Kirchhoff's junction rule implies currents at each junction equal to total current flowing through a battery



In an electric circuit, a source of voltage generates a **current** I , which is the flow of charge per time in a conductor (ie, wire, bulb, resistor). In addition, at a junction (where current splits or joins together), **Kirchhoff's junction rule** states that the sum of current flow entering the junction is equal to the sum of current leaving the junction:

$$I_{\text{enter}} = I_{\text{exit}}$$

Moreover, **Ohm's law** states that voltage V is equal to the product of resistance R and current:

$$V = IR$$

In this question, applying Kirchhoff's junction rule to the **parallel circuit** implies that the two measured currents must add up to the current flowing through the battery. The current flowing through the battery is calculated using Ohm's law. The battery voltage is 10 V, and the equivalent resistance of the circuit is 2 MΩ. Inserting these values into Ohm's law implies:

$$10 \text{ V} = I \cdot (2 \text{ M} \Omega)$$

$$10 \text{ V} = I \cdot (2 \times 10^6 \Omega)$$

Solving for I yields:

$$I = \frac{10 \text{ V}}{2 \times 10^6 \Omega} = 5 \times 10^{-6} \text{ A}$$

Thus, the current through the battery is 5 μA. Because the sum of the measured currents is 4.98 μA, and 5 μA is within the measurement uncertainty range of the experiment, the values are perfectly acceptable.

(Choice B) The values are smaller than the current flowing through the battery, but smaller currents in each branch are expected due to Kirchhoff's junction rule.

(Choice C) The sum of the measured current values is consistent with Kirchhoff's junction rule.

(Choice D) Measurement uncertainty arises due to limitations of the instrument being used. Uncertainties can be much larger than 1% of the measured value and still be valid.

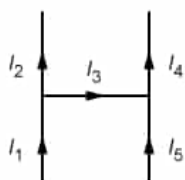
Educational objective:

Kirchhoff's junction rule states that the total current entering a junction must equal the total sum of each individual current leaving the junction. Current values measured in experiments are subject to uncertainty.

Time spent: 0 sec
Subject:
Physics

QID: 404044
System:
Electrostatics and Circuits

A segment of a circuit contains five wires carrying different currents labeled I_1 through I_5 , as shown.

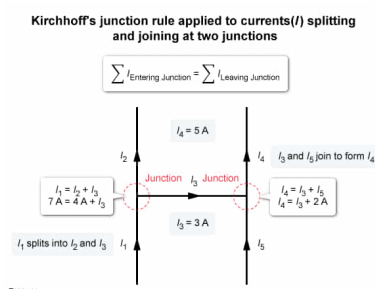


The directions of the currents are given in the figure. The values of I_1 , I_2 , and I_5 are 7 A, 4 A, and 2 A, respectively. What is the value of the current labeled I_4 ?

- ☐ A. 2 A [5%]
☐ B. 3 A [7%]
☐ C. 4 A [8%]
☒ D. 5 A [78%]

Correct
 77% answered correctly

Explanation:



In an electric circuit, **current** I is the flow of charge per time in a conductor (ie, wire, bulb, resistor). In addition, at a junction (where current splits or joins together in a circuit), **Kirchhoff's junction rule** follows from **conservation of charge**, implying that the sum of current entering the junction is equal to the sum of current leaving the junction:

$$\sum I_{\text{Entering Junction}} = \sum I_{\text{Leaving Junction}}$$

In this question, a circuit with two junctions is shown, along with the directions and values of the currents in each branch. Current I_1 splits at the left junction into currents I_2 and I_3 . Thus, Kirchhoff's junction rule implies:

$$I_1 = I_2 + I_3$$

Similarly, currents I_3 and I_5 enter the right junction and combine to form current I_4 . Therefore, it follows that:

$$I_3 + I_5 = I_4$$

Substituting the values of I_1 , I_2 , and I_5 into these equations gives the following:

$$7 \text{ A} = 4 \text{ A} + I_3$$

$$I_3 + 2 \text{ A} = I_4$$

Therefore, solving for I_3 in the first equation yields 3 A, which then can be plugged into the second equation to give $I_4 = 5 \text{ A}$.

(Choice A) 2 A is the difference between the values of I_3 and I_4 .

(Choice B) 3 A is the value of I_3 , not I_4 .

(Choice C) This answer is the value of I_2 , which could be confused with I_4 because both currents are pointing in the same direction in the upper part of the circuit diagram.

Educational objective:

Kirchhoff's junction rule states that charge must be conserved when currents split or join at junctions. Therefore, the total current entering a junction must equal the sum of each individual current leaving the junction.